claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a22393bb179597338.jsonl`
- SHA-256: `a3f37fc745ebbcd8843fc4f5527b58ffa3af3f16277ad36b1286346d06909452`
- Source modified: `2026-06-16T03:24:10+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_3d536535-241`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`

Transcript

1. user (2026-06-16T03:22:40.455Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):
- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).
- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.
- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RLOOCV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"An ablation on THUMOS14 isolates the contribution of each component: a convolutional baseline pyramid scores 62.1% avg mAP; replacing it with the SGP layer raises this to 68.3% (+6.2), and adding the Trident-head on top yields 69.3% (a further +1.0 over the instant-level regression head) ? quantifying that the relative-boundary head adds a measurable, separable boundary-localization gain."

Source: https://openaccess.thecvf.com/content/CVPR2023/papers/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.pdf (primary)
Supporting quote: "compared with the baseline model we implement (Row 1), the SGP layer brings a 6.2% absolute improvement in the average mAP... we compare our Trident-head with the normal instant-level regression head"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:22:43.593Z)

I'll verify this claim against the primary source (the TriDet CVPR 2023 paper). The claim makes specific numerical assertions about an ablation table, so I need to check the exact numbers.

3. assistant (2026-06-16T03:22:46.211Z)

I'll fetch the TriDet paper's ablation table to verify the specific numbers in the claim.

4. user (2026-06-16T03:22:49.106Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

5. assistant (2026-06-16T03:22:51.533Z)

The CVF site blocks the fetch. Let me try the arXiv version and search for the ablation numbers.

6. user (2026-06-16T03:22:55.693Z)

I cannot locate the specific ablation study details you're requesting in the provided webpage content. The page shows only the arXiv abstract and metadata for the TriDet paper (arXiv:2303.07347), which states that the method achieves "69.3% average mAP on THUMOS14" but does not include the detailed ablation table comparing baseline Conv, +SGP, +Trident-head components, or the quantitative comparison between the Trident-head and instant-level regression approaches.

To access those specific numerical results, you would need to view the full PDF of the paper itself.

7. user (2026-06-16T03:22:59.000Z)

Web search results for query: "TriDet CVPR 2023 ablation SGP layer 6.2% average mAP Trident-head THUMOS14 baseline"

Links: [{"title":"TriDet - CVPR 2023 Open Access Repository","url":"https://openaccess.thecvf.com/content/CVPR2023/html/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.html"},{"title":"TriDet: Temporal Action Detection with Relative Boundary Modeling","url":"https://arxiv.org/pdf/2303.07347"},{"title":"TriDet: Temporal Action Detection","url":"https://www.emergentmind.com/papers/2303.07347"},{"title":"[2303.07347] TriDet: Temporal Action Detection with Relative Boundary Modeling","url":"https://arxiv.org/abs/2303.07347"},{"title":"Noname manuscript No. (will be inserted by the editor)","url":"https://arxiv.org/pdf/2309.05590"},{"title":"[Quick Review] TriDet: Temporal Action Detection with Relative Boundary Modeling","url":"https://liner.com/review/tridet-temporal-action-detection-with-relative-boundary-modeling"},{"title":"TriDet: Temporal Action Detection with Relative Boundary Modeling | IEEE Conference Publication | IEEE Xplore","url":"https://ieeexplore.ieee.org/document/10203543/"},{"title":"TriDet: Temporal Action Detection with Relative Boundary Modeling Dingfeng Shi?","url":"https://openaccess.thecvf.com/content/CVPR2023/papers/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.pdf"},{"title":"CVPR Poster TriDet: Temporal Action Detection With Relative Boundary Modeling","url":"https://cvpr.thecvf.com/virtual/2023/poster/21956"}]

I found the information you're looking for! Compared with the baseline model, the SGP layer brings a 6.2% absolute improvement in the average mAP on THUMOS14.

This finding comes from the ablation study section of the TriDet paper presented at CVPR 2023. The study demonstrates the effectiveness of the proposed components in TriDet: SGP layer and Trident-head. The paper shows that benefiting from the Trident-head and the SGP-based feature pyramid, TriDet achieves state-of-the-art performance on three challenging benchmarks: THUMOS14, HACS and EPIC-KITCHEN 100.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-06-16T03:23:02.846Z)

The 6.2% SGP improvement is confirmed by search. Now I need the exact baseline number (62.1%), the +SGP number (68.3%), the +Trident number (69.3%), and the +1.0 Trident-head gain over instant-level regression head. Let me fetch the arXiv PDF directly to read the ablation table.

9. user (2026-06-16T03:23:09.004Z)

I cannot find an ablation table on THUMOS14 in the provided PDF content. The document appears to be a research paper in compressed/binary format, and while I can see references to figures and streams (intro.pdf, cosine.pdf, framework.pdf), the actual ablation study table with the specific mAP numbers you're asking about is not accessible in the readable text portions of this PDF.

To obtain the exact average mAP numbers for:
1. Baseline convolutional model
2. Baseline + SGP layer
3. Baseline + SGP + Trident-head
4. Instant-level regression head comparison

You would need to access the paper's results section directly, either through the original PDF's rendered version or the paper's supplementary materials.

[Binary content (application/pdf, 1.4MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579910495-xf7r2b.pdf]

10. assistant (2026-06-16T03:23:11.873Z)

The PDF was saved locally. Let me read it directly to find the exact ablation table.

11. user (2026-06-16T03:23:13.108Z)

pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)

12. assistant (2026-06-16T03:23:14.919Z)

Let me try extracting the text from the PDF using Python instead.

13. user (2026-06-16T03:23:19.839Z)

=== KW: 62.1 ===

=== KW: 68.3 ===

=== KW: 69.3 ===

=== KW: instant ===

=== KW: Trident-head ===

=== KW: SGP layer ===

=== KW: baseline ===

=== KW: Baseline ===

14. assistant (2026-06-16T03:23:22.454Z)

Empty results may mean pypdf isn't installed or extraction failed silently. Let me check.

15. user (2026-06-16T03:23:25.311Z)

```
HAVE pdfplumber
HAVE pypdf
NO PyPDF2
HAVE fitz
```

16. assistant (2026-06-16T03:23:28.586Z)

```
pypdf is available but extraction returned nothing. Let me try fitz (PyMuPDF) which usually
extracts better.
```

17. user (2026-06-16T03:23:32.011Z)

```
TOTAL LEN 60502
=== KW: 62.1 ===
'77.3 65.2 40.0 62.1'

=== KW: 68.3 ===
'68.3 62.3 51.9 38.8 23.7'
'68.3 63.8 54.3 41.8 26.2'
'83.6 71.7 45.8 68.3'
'83.6 45.8 68.3 14.5'
'68.3 68.5 '
'68.3 68.5 '
'68.3'
'68.3'

=== KW: 69.3 ===
'69.3% on THUMOS14, outperforming the previous best by'
'69.3'
'method achieves an average mAP of 69.3%, outperform-'
'83.6 72.9 47.4 69.3'
'83.6 47.4 69.3 14.5'
'69.3 '
'69.3'
'69.3'
'69.3'

=== KW: instant ===
'and end instants and corresponding action categories from'
'regress the boundaries based on a single instant, potentially with'
'tion at each instant. As another line of work, they directly'
'regress the boundaries based on a single instant [33,49], po-'
'consider the relation between adjacent instants (e.g. the rel-'
'instant and the video-level average feature for self-attention and'
'ination of features in each instant and capture temporal in-'
'match the start and end instants according to the prediction'
'where T corresponds to the number of instants, and Ki seg-'
'start instant sk, the end instant ek and the corresponding'
'angle between features within the instant features.'
'tion, the dense pair-wise calculation (between instant fea-'
'an instant-level branch and a window-level branch. In the'
'instant-level branch, we aim to increase the feature discrim-'
'inability between action and non-action instant by enlarging'
'instant. At the instant t, the predicted boundary response in neigh-'
'responding to the instant t, which is further estimated as the rel-'
'temporal length [49] of the action at each instant of the'
'the relative probability of being a boundary for each instant'
'and Fe characterize the response value for each instant as'
'the probability that each instant (in its set of bins) serves as'
'a boundary when the instant t is the midpoint of an action.'
'of the left adjacent bin set of instant t and the center off-'
'sets predicted by instant t only, respectively, and ePst is the'
'instant as a start of the action within the bin set. Then, the'
```

'distance between the instant t and the start instant of action'
'Formally, for an instant in the l-th feature level tl,'
'egories: the prediction based on sampling instants in seg-'
'instant. The first category predicts the boundary accord-'
'detailed information at each instant.'
'gory directly predicts the distance between an instant and'
'its corresponding boundary based on the instant-level fea-'
'adjacent instants into account. The proposed Trident-head'
'tion. The output of each instant t in feature pyramid layer l'
'used to reweight the classification loss at each instant, such'
'that instants with better regression (i.e. of higher quality)'
'positive samples. Namely, the instants around the center of'
'Inference. At inference time, the instants with classifica-'
'a normal instant-level regression head.'
'we compare our Trident-head with the normal instant-level'
'each instant. We can see that the Trident-head improves the'
'tures at each instant indistinguishable, resulting in inaccu-'
'accurately predicts the start and end instant of the action.'

=== KW: Trident-head ===
'lem, we propose a novel Trident-head to model the action'
'Trident-head and the SGP-based feature pyramid, TriDet'
'named Trident-head tailored for action boundary localiza-'
'Trident-head models the action boundary via an estimated'
'Apart from the Trident-head, in this work, the proposed'
'Trident-head'
'Trident-head. The Trident-head estimates the boundary offset based on a relative distribution predicted by three branche'
'oriented Trident-head. First, the video features are extracted'
'Trident-head (Section 3.3). We elaborate on the proposed'

18. assistant (2026-06-16T03:23:36.274Z)

Now I have key data. I see `62.1` appears as `77.3 65.2 40.0 62.1` (this is an avg mAP column). And `69.3` appears in ablation context `83.6 72.9 47.4 69.3`. I need to find the specific ablation table that shows the progression baseline ? +SGP ? +Trident. Let me extract more context around the ablation table and the "6.2" / "instant-level regression head" comparison.

19. user (2026-06-16T03:23:42.632Z)

==== KW: 6.2 ====
68.3 63.8 54.3 41.8 26.2

6.2

54.1 36.2
---
we implement (Row 1), the SGP layer brings a 6.2% abso-

83.4 46.2

56.2 38.4 11.2

82.8 71.8 46.2

56.2 38.5 11.1

83.6 72.3 46.2
---
the CNN baseline (about 6.2%) but is still behind the Trans-

==== KW: absolute improvement ====

==== KW: 62.1 ====

77.3 65.2 40.0 62.1

==== KW: Trident-head improves ====
each instant. We can see that the Trident-head improves the

==== KW: instant-level regression head ====
a normal instant-level regression head.

==== KW: SGP layer brings ====
we implement (Row 1), the SGP layer brings a 6.2% abso-
---

20. assistant (2026-06-16T03:23:45.537Z)

I have the "6.2% absolute improvement" passage confirmed. Now I need to pin down the exact ablation table rows. The claim says baseline=62.1, +SGP=68.3, +Trident=69.3. Let me extract the full ablation section text with surrounding context, particularly the table near "Row 1" and the instant-level regression comparison.

21. user (2026-06-16T03:23:48.877Z)

--- CONTEXT AROUND SGP layer brings (chars) ---
re
which is not available now.
4.3. Ablation Study
In this section, we mainly conduct the ablation studies
on the THUMOS14 dataset.
Main components analysis.
We demonstrate the effec-
tiveness of our proposed components in TriDet: SGP layer
and Trident-head. To verify the effectiveness of our SGP
layer, we use a baseline feature pyramid used by [21, 49]
to replace our SGP layer. The baseline consists of two 1D-
convolutional layers and shortcut. The window size of con-
volutional layers is set to 3 and the number of channels of
the intermediate feature is set to the same dimension as the
Table 5. Analysis of the Effectiveness of three main components
on THUMOS14.
Method SA SGP Trident
0.3
0.5
0.7
Avg.
1
77.3 65.2 40.0 62.1
2
?
82.1 71.0 43.9 66.8
3
?
83.6 71.7 45.8 68.3
4
?
?
83.6 72.9 47.4 69.3
Table 6. Analysis of computation cost on THUMOS14. Main: All
parts of the model except the detection head. *: Our method with
a normal instant-level regression head.
Method
mAP
GMACs
Latency

(ms)
0.3
0.7 Avg. Main Head All
ActionFormer 82.1 43.9 66.8 30.8
14.4 45.3
224
TriDet?
83.6 45.8 68.3 14.5
14.4 28.9
145
TriDet
83.6 47.4 69.3 14.5
29.1 43.7
167
intermediate dimension in the FFN in our SGP layer. All
other hyperparameters (e.g. number of the pyramid layers,
etc.) are set to the same as our framework.
As depicted in Tab. 5, compared with the baseline model
we implement (Row 1), the SGP layer brings a 6.2% abso-
lute improvement in the average mAP. Secondly, we com-
pare the SGP with the previous state-of-the-art method, Ac-
tionFormer, which adopts a self-attention mechanism in a
sliding window behavior [4] with window size 7 (Row 2).
We can see our SGP layer still has 1.5% improvement in
average mAP, demonstrating that the convolutional network
can also have excellent performance in TAD task. Besides,
we compare our Trident-head with the normal instant-level
regression head, which regresses the boundary distance for
each instant. We can see that the Trident-head improves the
average mAP by 1.0%, and the mAP improvement is more
obvious in the case of high IoU threshold (e.g. 1.6% average
mAP improvement in IoU 0.7).
Computational complexity.
We compare the computa-
tional complexity and latency of TriDet with the recent Ac-
tionFormer [49], which brings a large improvement to TAD
by introducing the Transformer-based feature pyramid.
As shown in Tab. 6, we divide the detector into two parts:
the main architecture and the detection heads (e.g. classifi-
cation head and regression head). We report the GMACs
for each part and the inference latency (average over five
times) on THUMOS14 dataset using an input with the shape
2304 × 2048, following the [49]. We also report our results
using the Trident-head and the normal regression head, re-
spectively. First, f

22. assistant (2026-06-16T03:23:59.154Z)

I now have the complete ablation table (Table 5) verbatim from the primary source. Let me verify each number in the claim against it.

Table 5 (THUMOS14 main components), columns = 0.3 / 0.5 / 0.7 / Avg:
- Row 1 (baseline conv, no SA/SGP/Trident): 77.3 / 65.2 / 40.0 / **62.1**
- Row 2 (SA / ActionFormer self-attention): 82.1 / 71.0 / 43.9 / 66.8
- Row 3 (SGP): 83.6 / 71.7 / 45.8 / **68.3**
- Row 4 (SGP + Trident): 83.6 / 72.9 / 47.4 / **69.3**

Now checking the claim point-by-point:

1. "convolutional baseline pyramid scores 62.1% avg mAP" ? **CONFIRMED** (Row 1 = 62.1).
2. "replacing it with the SGP layer raises this to 68.3% (+6.2)" ? **CONFIRMED** (Row 3 = 68.3; text: "the SGP layer brings a 6.2% absolute improvement in the average mAP"; 68.3 ? 62.1 = 6.2).
3. "adding the Trident-head on top yields 69.3% (a further +1.0 over the instant-level regression head)" ? **CONFIRMED** (Row 4 = 69.3; text: "we compare our Trident-head with the normal instant-level regression head... the Trident-head improves the average mAP by 1.0%").

69.3 ? 68.3 = 1.0, and Table 6's TriDet* (instant-level regression head) = 68.3 avg, TriDet (Trident-head) = 69.3 avg, confirming the +1.0 is over the instant-level regression head.

Every number and the interpretation ("relative-boundary head adds a measurable, separable boundary-localization gain") are directly supported by the primary source. The 1.0 gain being concentrated at high IoU (1.6% at IoU 0.7) further supports the "boundary-localization gain" characterization. The only minor imprecision: the claim labels Row 1 as "convolutional baseline pyramid" ? the paper describes it as a baseline of "two 1D-convolutional layers and shortcut," which is accurate. No contradictions found. This is a primary CVPR 2023 source; the claim is current and well-supported.

23. user (2026-06-16T03:24:07.384Z)

Structured output provided successfully

24. assistant (2026-06-16T03:24:10.481Z)

Verified the claim against the primary source. All numbers (62.1 baseline, 68.3 +SGP/+6.2, 69.3 +Trident/+1.0 over instant-level regression head) match TriDet Table 5/Table 6 and the paper's own stated deltas exactly. Claim is well-supported by the primary CVPR 2023 source ? **refuted=false, high confidence**.